

THAPAR STEPUP

Turning Campus Walking Into a Daily Game

A gamified walking platform that turns everyday campus movement into streaks, challenges and rewards.



THE PROBLEM

Students Walk Across Campus. But There's No Motivation.



Hours of Sitting

Classes, labs and libraries keep students seated most of the day.



Low Motivation

Nothing pushes students to move beyond the bare minimum.



Generic Fitness Apps

Built for anyone, everywhere — not for campus life.



No Real Competition

Missing the social edge that keeps people coming back.



No Reward for Consistency

Nothing to gain by showing up and walking every day.



6+

HOURS SPENT SITTING, DAILY

“Students already walk around campus — we turn those steps into motivation.”

SOLUTION

Meet Thapar StepUp

One simple idea: turn every walk across campus into progress.



BUILT AROUND REAL CAMPUS ROUTES

Hostel → Academic Block

Hostel → Library

Library → Sports Complex

Academic Block → Cafeteria

HOW THE GAMIFICATION WORKS

Every Step Becomes Progress



EXAMPLES

-  30-Day Streak
-  Top 10 Campus Walker
-  10K Steps Challenge
-  Hostel Champion
-  5,000 XP

WHY THAPAR STEPUP

More Than a Step Counter

GENERIC FITNESS APPS

- Individual tracking only
- Generic, one-size goals
- No campus context
- Limited student competition
- No campus-specific challenges

- Campus-specific tracking
- Hostel & class leaderboards
- Real campus routes & milestones
- Daily streaks & challenges
- Rewards + social motivation

IMPACT & VALIDATION

How We Measure Success

Daily Active Users

Average Daily Steps



**7 / 30-Day Streak
Retention**

Challenge Completion Rate

**Leaderboard
Participation**

**Average Weekly
Walking**

VALIDATION PLAN

Prototype



**Pilot with
Students**



**Collect
Data**



**Measure
Engagement**



Iterate



FEASIBILITY & RISKS

Built to Be Real

FEASIBILITY

- Smartphone step sensors
- Campus geofencing where required
- Cloud / database backend
- Gamification logic engine
- Simple student onboarding

RISKS

- GPS / step accuracy
- Battery usage
- Privacy concerns
- Cheating / fake steps
- Long-term engagement

Make every walk across campus count.

SmartGate — ANPR Based Intelligent Vehicle Access

DISHA GOYAL

Problem Statement

- Every registered vehicle — faculty, staff, or student — is issued a colored sticker that determines their access level on campus.
- During peak hours — 8 to 9 AM and 5 to 6 PM — hundreds of vehicles pass simultaneously, making accurate checking impossible.
- Stickers fade over time due to sun exposure and weather. They can be misused — a student can enter a staff-only zone if the guard is not attentive.
- There is no digital record of which vehicles entered, at what time, through which gate, or how long they stayed.
- The core problem is simple — Thapar has no intelligent, automated, and digitized system to manage vehicle entry, access control, and campus security.

Automatic Number Plate Recognition — Built for Thapar

ANPR cameras read every plate in under **2 seconds**, match it against a registered database, and decide access automatically.

- Real-Time Plate Recognition

Camera captures plate → OCR reads it → access decision returned in under 2 seconds.

- Five Access Categories

Faculty, staff, student, pre-registered visitor, and unregistered — each with defined permissions.

- Pre-Visitor Registration

Students and faculty register guest plates in advance. Guest enters automatically; host is notified.

Even unreadable plates are photographed, timestamped, and flagged — **no vehicle enters anonymously.**

Smart Solutions Built Into the System

- **Zero Digital Records** — Every entry captured with visitor name, time, date, and gate number in a structured, searchable database.
- **No Incident Trail** — Time-stamped, searchable records of every entry/exit event, so incidents can be traced in seconds instead of scrubbing hours of raw CCTV footage.
- **Visitor & Vehicle Tracking** — Complete record of every state vehicle's visits, including frequency, purpose, and duration on campus.

Tech Stack, Implementation & Proven Feasibility

Technology Stack

- **Camera & OCR:** Python, OpenCV, EasyOCR — Indian plates in real time
- **Backend:** FastAPI — queries database, applies rules, returns decision under 2 seconds
- **Database:** PostgreSQL — plates, timestamps, images, gates, decisions
- **Dashboard:** React — live entries, flagged vehicles, PDF reports
- **Pre-Visitor Portal:** Mobile-friendly web — no app needed

Loophole??!!

Novelty — Government Vahan API

- Plate number is read by camera — one API call to government Vahan database instantly returns vehicle class, owner details, blacklist status, fuel type, insurance status etc.
- Vehicle class decides time automatically — Motor Cab gets 1 hour, Goods Carrier gets 6 hours, 2 wheelers gets 2-3 hours depending on requirements.
- Blacklisted or suspended vehicles trigger instant guard alert — no vehicle enters anonymously even if flagged by RTO

CampusPrep

**Campus Specific Placement Experience and
Preparation Platform**

Problem Statement

- Students preparing for campus placements often depend on seniors, WhatsApp groups, friends, PDFs, and general platforms to find interview experiences.
- The information is scattered and difficult to compare, especially when students are looking for experiences related to a specific company, role, year, or interview round.
- General placement platforms may provide interview information, but they do not necessarily reflect the recruitment process followed by a company on a particular campus.
- Students therefore spend significant time searching for relevant information and may still not get a clear idea of what to prepare for a particular campus opportunity.
- The core problem is simple — there is no single, structured platform that brings together campus-specific placement experiences in an organized and easily searchable form.

What Will We Do Better?

- Cover all types of companies visiting campus from large MNCs to smaller companies and startups.
- Organize experiences by company, role, year and interview round.
- Add filters for difficulty level, type of questions, technical/HR rounds and topics asked.
- Make it easy to search for a specific company or role.
- Keep the information structured and easy to compare.

Proposed System

Student

- Search company/role
- Filter by year/round
- View structured experiences
- Get preparation insights

Contributor

- Submit experience
- Select company, role, year & rounds
- Add generalized experience
- Submit anonymously

Reporting

- Report inappropriate or incorrect information
- Report privacy/confidentiality concerns
- Flag potentially harmful content

How Will We Validate It?

- Talk to students who have already gone through placements.
- Find out what information they needed and where they got it from.
- Let students try our platform with a few real company examples.
- Check if they are able to find the information they need without much effort.
- Take their feedback and improve the platform accordingly.

Feasibility & Risk

Feasibility

- Web-based application.
- Can be developed using standard web technologies.
- Can start with a limited number of companies and expand.
- Data can be collected through student contributions.

Risks & Mitigation

- **Low number of submissions** → Make contribution simple and anonymous.
- **Incorrect information** → Reporting option and periodic review.
- **Confidential information** → Clear submission guidelines and content review.
- **Privacy concerns** → Collect minimal personal information.
- **Incomplete information** → Clearly show available experiences and their year/role.

Why Select CampusPrep?

- Solves a real problem faced by students during placements.
- Helps students learn from the experiences of students who went through the process before them.
- Makes it easier to identify what to prepare for a particular role.
- Information can be updated every year, so the platform keeps growing.
- Gives students a single starting point instead of asking multiple people or searching different platforms which saves a lot of time.
- One structured place for all campus placement experiences.